EXAMPLE ACI 318-19 Wall-001

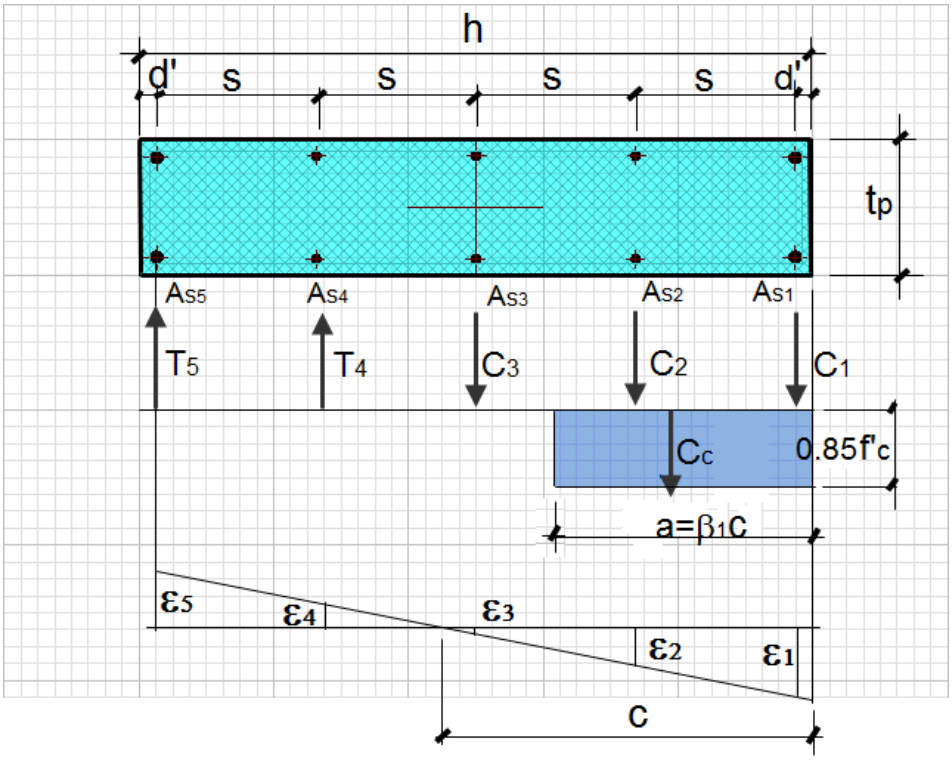
P-M INTERACTION CHECK FOR WALL

EXAMPLE DESCRIPTION

The Demand/Capacity ratio for a given axial loading and moment is tested in this example.

A reinforced concrete wall is subjected to factored axial load P_u = 735 k and moments M_{uy} = 1,503 k-ft. This wall is reinforced with two #9 bars at each end and #4 bars at 14.00 inches on the center of each face. The total area of reinforcement is 5.20 in^2. The design capacity ratio is checked by hand calculations, and results are compared with ETABS program results.

GEOMETRY, PROPERTIES AND LOADING



Material Properties	Section Properties	Design Properties
E = 3600 k/in^2	tb = 12 in	f'_c = 4 k/in^2
v = 0.2	h = 60 in	f_y = 60 k/in^2
G = 1500 k/in^2	As1= As5 = 2-#9 (2.00 in^2)	
	As2, As3, As4 = 2-#4 (0.40 in^2)	

TECHNICAL FEATURES OF ETABS TESTED

- Concrete wall flexural Demand/Capacity ratio

RESULTS COMPARISON

Independent results are hand calculated and compared with ETABS design check.

Output Parameter	ETABS	Independent	Percent Difference
Wall Demand/Capacity Ratio	1.01	1.00	1.00%

COMPUTER FILE: ACI 318-19 WALL-001

CONCLUSION

The ETABS results show an acceptable match with the independent results.

HAND CALCULATION

COLUMN STRENGTH UNDER COMPRESSION CONTROL

1) A value of $e = 24.54$ inch was determined using $e = M_u / P_u$ where M_u and P_u were taken from the ETABS test model. The values of M_u and P_u were large enough to produce a flexural D/C ratio very close to or equal to one. The depth to the neutral axis, *c*, was determined by iteration using an excel spreadsheet so that equations 1 and 2 below were equal.

2) From the equation of equilibrium:

$$P_n = C_c + C_s - T$$

where

$$C_c = 0.85 f_c' ab = 0.85 \bullet 4 \bullet 12a = 40.8a$$

$$C_s = A_1'\left(f_{s1} - 0.85f_c'\right) + A_2'\left(f_{s2} - 0.85f_c'\right) + A_3'\left(f_{s3} - 0.85f_c'\right)$$

$$T = A_{s4}f_{s4} + A_{s5}f_{s5}$$

$$P_{n1} = 40.8a + A_1'\left(f_{s1} - 0.85f_c'\right) + A_2'\left(f_{s2} - 0.85f_c'\right)A_1'\left(f_{s1} - 0.85f_c'\right) + A_3'\left(f_{s3} - 0.85f_c'\right) - A_{s4}f_{s4} - A_{s5}f_{s5} \quad \text{(Eqn. 1)}$$

3) Taking moments about A_{s5}:

$$P_{n2} = \frac{1}{e'}\left[C_c\left(d - \frac{a}{2}\right) + C_{s1}\left(d - d'\right) + C_{s2}\left(3s\right) - C_{s3}\left(2s\right) - T_{s4}\left(s\right)\right] \quad \text{(Eqn. 2)}$$

where $C_{s1} = A'\left(f_{s1} - 0.85f_c'\right)$; $C_{sn} = A_n'\left(f_{sn} - 0.85f_c'\right)$; $T_{sn} = f_{sn}A_{sn}$; and the bar strains are determined below. The plastic centroid is at the center of the section and $d'' = 28$ inch

$$e' = e + d'' = 24.54 + 28 = 52.54 \text{ inch.}$$

4) Using $c = 30.1$ inch (from iteration),

$$a = 0.85 \bullet 30.1 = 25.58 \text{ inches}$$

5) Assuming the extreme fiber strain equals 0.003 and *c*= 30.1 inch, the steel stresses and strains can be calculated. When the bar strain exceeds the yield strain, then $f_s = f_y$:

$$\varepsilon_{s1} = \left(\frac{c-d'}{c}\right)0.003 \quad = 0.0028; f_s = \varepsilon_s E \le F_y \;;\; f_{s1} = 60.00 \text{ ksi}$$

$$\varepsilon_{s2} = \left(\frac{c-s-d'}{c}\right)0.003 \quad = 0.0014 \qquad f_{s2} = 40.75 \text{ ksi}$$

$$\varepsilon_{s3} = \left(\frac{d-c-2s}{d-c}\right)\varepsilon_{s5} \quad = 0.0000 \qquad f_{s3} = 00.29 \text{ ksi}$$

$$\varepsilon_{s4} = \left(\frac{d-c-s}{d-c}\right)\varepsilon_{s5} \quad = 0.0014 \qquad f_{s4} = 40.20 \text{ ksi}$$

$$\varepsilon_{s5} = \left(\frac{d-c}{c}\right)0.003 \quad = 0.0028 \qquad f_{s5} = 60.00 \text{ ksi}$$

Substituting in Eqn. 1 and 2 and iterating the value of the neutral axis depth until the two equations are equal give

$P_{n1} = 1035$ k
$P_{n2} = 1035$ k

$M_n = P_n e = 1035(24.54)/12 = 2115$ k-ft

6) Determine ϕ based on maximum strain value ε_t.

$\varepsilon_t = 0.0028$, $\varepsilon_{ty} = \dfrac{60{,}000}{29{,}000{,}000} = 0.0020689$

For $\varepsilon_{ty} < \varepsilon_t < \varepsilon_{ty} + 0.003$:

$$\phi = \phi_c + (\phi_t - \phi_c)\left(\frac{\varepsilon_t - \varepsilon_{ty}}{0.003}\right) = 0.65 + (0.90 - 0.65)\left(\frac{0.0028 - 0.0020689}{0.003}\right) = 0.7109$$

7) Calculate ϕP_n and ϕM_n.

$\phi P_n = 0.7109(1035) = 735$ kips
$\phi M_n = 0.7109(2115) = 1503$ k-ft.

8) Calculate the D/C ratio.

$$\text{D/C} = \frac{P_u}{\phi P_n} = \frac{735}{735} = 1.00$$

$$\text{D/C} = \frac{M_u}{\phi M_n} = \frac{1503}{1503} = 1.00$$